

QISKIT FALL FEST 2026

UNIVERSITY of ENGINEERING and TECHNOLOGY LAHORE, NAROWAL CAMPUS

Weekly Planning Schedule

Event Schedule • Day-wise Speakers and Topics

9	2–18 Oct 2026	2.5 Hours
Event Days	Event Window	Speaker Session / Day

1. Event Schedule

#	Date	Time (Tentative)	Day	Theme	Guest Speaker(s)
1	2 October 2026	6:00 pm to 7:15 pm PKT 7:30 pm to 8:45 pm PKT	FRI	Foundations — Qubits, Gates & Superposition	Dustin Rippe; Dr. Fatima Bint e Munir
2	3 October 2026	6:00 pm to 8:30 pm PKT	SAT	Measurement, Circuits & Qiskit Patterns	Mehar Ali
3	4 October 2026	6:00 pm to 8:30 pm PKT	SUN	Multi-Qubit Circuits & Entanglement	Dr. Fatima Bint e Munir
4	9 October 2026	6:00 pm to 8:30 pm PKT	FRI	Circuit Optimization & Transpilation	Dustin Rippe
5	10 October 2026	6:00 pm to 7:15 pm PKT 7:30 pm to 8:45 pm PKT	SAT	Noise, Backends & Real Hardware	Dustin Rippe; Muhammad Khizar
6	11 October 2026	6:00 pm to 7:45 pm PKT 8:00 pm to 8:45 pm PKT	SUN	Dynamic Circuits & Parameterized Circuits	Dr. Fatima Bint e Munir Fatima Shahzadi
7	15 October 2026	12:00 pm to 4:00 pm PKT	THU	LAB 1	Instructor: Noor Ul Ain
8	16 October 2026	9:00 am to 01:00 pm PKT	FRI	LAB 2	Instructor: Noor Ul Ain
9	18 October 2026	2:00 pm to 6:00 pm PKT	SUN	Hackathon	All Guest Speakers

Standard speaker session: 2.5 hours per event day

2. Day-wise Speakers & Topics

Day 1 — Foundations: Qubits, Gates & Superposition

Dustin

- Introduction to Quantum Computing
- Python environment setup for Qiskit 2.x
- Vectors, matrices and inner products
- Qiskit Statevector and Operator

Dr. Fatima Bint e Munir

- Single-qubit gates: X, H, Z, Y, S, T
- Superposition

Day 2 — Measurement, Circuits & Qiskit Patterns

Mehar Ali

- Measurement
- Bloch sphere visualization
- Introduction to state tomography
- Qiskit Patterns: Map, Optimize, Execute, Post-process
- Sampler Primitive
- Estimator Primitive

Day 3 — Multi-Qubit Circuits & Entanglement

Dr. Fatima Bint e Munir

- Multi-qubit state representation
- Tensor products
- Entanglement and lab demonstration
- Bell states
- GHZ states

Day 4 — Circuit Optimization & Transpilation

Dustin

- Circuit depth: what it is and why it matters
- Reducing depth: fan-out vs. chain; recursive fan-out
- Transpilation deep dive
- GHZ optimization: fan-out, chain, recursive
- Heavy-Hex topology

Day 5 — Noise, Backends & Real Hardware

Dustin

- Noise in quantum computers: types of noise and sources
- Noisy simulation
- Noise models: depolarization, Pauli, thermal

Muhammad Khizar

- Quantum microprocessor (advanced topic)
- Backend properties: T1, T2, gate errors, readout errors

Day 6 — Dynamic Circuits & Parameterized Circuits

Dr. Fatima Bint e Munir

- Classical registers
- Mid-circuit measurement
- Conditional logic
- Dynamic circuits: step-by-step explanation

Fatima Shahzadi

- Parameter and ParameterVector
- Real-world quantum applications

Day 7 — Hackathon & Project Day

All Speakers

- Hackathon tips & tricks
- Hackathon project work & presentations